

Engine Failure Prediction — Project Report

RUL Estimation with NASA C-MAPSS | LightGBM

Project Summary

This project focuses on developing a machine learning-based Remaining Useful Life (RUL) prediction system for turbofan engines using NASA's C-MAPSS dataset. The main objective is to enable predictive maintenance by estimating how many operational cycles remain before an engine is likely to fail. This foresight allows maintenance teams to make informed, timely decisions—balancing safety, cost, and operational efficiency.

Data Overview

- **Train Set:** Complete run-to-failure data for 100 engines
- **Test Set:** Partial life data up to a point before failure
- **RUL File:** Remaining Useful Life for each engine in the test set

Each record includes:

- Engine ID
- Cycle number
- 3 operational settings
- 21 sensor readings (columns 6–26)

Feature Engineering

- Extracted maximum cycle per engine to define the end-of-life point (EOL)
- Created the Remaining Useful Life (RUL) target as: **$RUL = EOL - cycle_time$**
- Selected sensors with known degradation trends; dropped low-variance/non-informative sensor columns and operational settings after inspection
- Imputed missing sensor readings (mean strategy) and scaled all features with StandardScaler. This fitted imputer+scaler is saved as its own artifact and applied identically wherever the model is used, so training and the deployed dashboard can never disagree on preprocessing.
- A single shared feature-engineering module defines the raw column list, the dropped-feature list, and the inference logic once, and is imported by both the training notebook and the monitoring dashboard—eliminating the risk of the two drifting out of sync with each other.
- Evaluated IQR-based outlier removal on the degradation sensors and rejected it: since these sensors are expected to drift as an engine approaches failure, the filter disproportionately removed near-failure training rows—exactly the examples the model most needs to learn from—which measurably hurt recall on the highest-risk class in testing. No outlier filtering is applied to the sensor data.

Model Selection and Justification

Multiple models were evaluated (3-fold cross-validated RMSE):

Model	RMSE
Support Vector Regressor	38.72
Random Forest Regressor	36.48
LightGBM	36.24

Selected Model: LightGBM

Reasons for selection:

- Best performance score during experimentation
- High performance on tabular time-series data
- Built-in feature importance
- Fast training and evaluation time

Performance Summary

- **MAE:** 17.93
- **RMSE:** 24.68
- Top predictive features: cycle_time, NRC (corrected core speed), Nc (physical core speed), phi (fuel/air ratio), Ps30 (static pressure)
- Performed best on engines with longer run histories
- Slight **overestimation** of RUL for early-failure cases (engines that fail before ~100 cycles)

Maintenance Planning Use Case

The RUL prediction output can be directly integrated into a maintenance system. The RUL prediction model enables a data-driven maintenance strategy, transitioning from reactive or schedule-based approaches to predictive maintenance.

System Integration:

- Daily or real-time ingestion of sensor data
- Automatic risk classification based on RUL predictions

Predicted RUL	Recommended Maintenance Action
≤ 15 cycles	Immediate Repair
16 – 47 cycles*	Schedule Inspection
> 47 cycles*	Normal Operation Monitoring

* The 15-cycle Repair cutoff is fixed. The 47-cycle boundary shown above is the threshold used during evaluation; in the deployed dashboard it is an adjustable slider (default 30, range 10–70) so operators can tune it to their own risk tolerance.

- Maintenance decision support dashboard
- Scheduled intervention planning

Benefits:

- Failures are detected before they occur, minimizing unscheduled outages
- Eliminates unnecessary early replacements while avoiding catastrophic failures
- Supports prioritization of engine maintenance through data-driven risk ranking

Conclusion and Recommendations

The LightGBM-based RUL prediction system performs reliably, especially for engines with longer histories, enabling proactive, condition-based maintenance. By forecasting failures in advance, it can help reduce downtime, extend asset life, and optimize resource planning.

The model does overestimate RUL in short-lived engines, which can be mitigated by adjusting thresholds (e.g., monitor at ≤ 47 cycles, repair at ≤ 15), allowing teams to balance early warnings with operational risk.

Limitations and Future Work

- **Class Imbalance:** Adjust training to better capture short-lived engine behavior
- **Early Signals:** Add features that reflect early degradation trends
- **Temporal Learning:** Explore LSTM or Transformer models for sequence data
- **Sensor Noise:** Use smoothing or filtering to address sensor instability
- **Dynamic Thresholds:** Tie maintenance flags to operational context and cost-risk tradeoffs
- **Optimization:** Integrate cost-aware planning for greater operational impact
- **Engine-Grouped Validation:** Re-split train/validation by engine ID rather than by row, so validation scores reflect generalization to genuinely unseen engines rather than partly interpolating within engines already seen in training